

HORIZON MEDICAL — 510(k) Premarket Notification

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510(k) Number: [To be assigned by FDA]

Referencia: 21 CFR Part 807 Subpart E

Cover Letter

[Date]

Document Mail Center (HFZ-401)
Center for Devices and Radiological Health
Food and Drug Administration
10903 New Hampshire Avenue
Silver Spring, MD 20993-0002

Re: 510(k) Premarket Notification for Horizon Medical Holter ECG System (HM-H100)

Dear Sir/Madam:

Enclosed please find a 510(k) Premarket Notification for the Horizon Medical Holter ECG System (HM-H100), an ambulatory electrocardiographic monitoring system with AI-assisted arrhythmia detection, manufactured by [Manufacturer Name].

This submission is being made pursuant to Section 510(k) of the Federal Food, Drug, and Cosmetic Act. The device is classified under 21 CFR 870.2340, Product Code DRX (Electrocardiograph, Ambulatory), as a Class II medical device.

The predicate device is the iRhythm Zio XT Patch System (K142497). Additional reference devices include BioTelemetry MCOT (K162604) and Cardiac Insight Cardea SOLO (K173437).

A User Fee has been submitted as required under MDUFA.

We certify that the information provided in this submission is truthful and accurate, and that no material fact has been omitted.

Please contact the undersigned with any questions regarding this submission.

Respectfully submitted,

[Name], [Title]

[Company Name]

[Address]

[Phone] | [Email]

Contact Person for FDA Communications: [Name], [Title]

Section 1: 510(k) Summary (21 CFR 807.92)

1.1 Submitter Information

Field	Detail
Submitter	[Company Name]
Address	[Full Address]
Contact Person	[Name, Title]
Phone	[Phone Number]
Email	[Email Address]
Date Prepared	[Date]

1.2 Device Information

Field	Detail
Device Trade Name	Horizon Medical Holter ECG System
Device Common Name	Ambulatory Electrocardiographic Monitoring System
Device Model	HM-H100
Regulatory Class	Class II
Product Code	DRX
Regulation Number	21 CFR 870.2340
Panel	Cardiovascular

1.3 Predicate Device

Field	Detail
Predicate Trade Name	Zio XT Patch System
Predicate 510(k) Number	K142497
Predicate Manufacturer	iRhythm Technologies, Inc.

1.4 Device Description Summary

The Horizon Medical Holter ECG System is a prescription-use ambulatory electrocardiographic monitoring system intended for continuous recording and analysis of ECG signals for up to 48 hours. The system consists of a wearable ECG acquisition device (3-channel, Ag/AgCl surface electrodes), a mobile application (iOS/Android), a cloud-based analysis backend incorporating an AI/ML-based arrhythmia detection algorithm (CNN-LSTM), and a clinical web dashboard.

1.5 Substantial Equivalence Summary

The Horizon Medical Holter ECG System has the same intended use as the predicate device (ambulatory ECG monitoring with arrhythmia detection). Technological differences (3-channel vs. 1-channel ECG, real-time BLE connectivity, CNN-LSTM algorithm) do not raise new questions of safety or effectiveness, as demonstrated by bench testing, software verification and validation, AI algorithm validation, biocompatibility evaluation, and usability testing.

Section 2: Indications for Use (FDA Form 3881)

Indications for Use Statement

510(k) Number: [To be assigned]

Device Name: Horizon Medical Holter ECG System

Indications for Use:

The Horizon Medical Holter ECG System is a prescription-use device intended for use by healthcare professionals for the continuous recording, storage, analysis, and display of ambulatory electrocardiographic (ECG) data from adult patients (≥ 18 years) for up to 48 hours.

The device records 3-channel ECG signals using surface electrodes placed on the patient's chest. The system includes an AI-based software function that provides automated detection and classification of cardiac arrhythmias, including atrial fibrillation, ventricular tachycardia, bradycardia, and premature complexes, as a clinical decision support tool.

The AI-based arrhythmia detection results are intended to assist, not replace, the clinical judgment of a qualified healthcare professional. All AI-generated results must be reviewed and confirmed by a physician before any clinical action is taken.

Prescription Use ☒

Over-The-Counter Use ☐

Section 3: Device Description

3.1 System Overview

[See detailed description in 00-CLASIFICACION-DISPOSITIVO.md, Section 1]

The system comprises:

1. **Wearable ECG Device (HM-H100):** Portable 3-channel ECG recorder with nRF52832 SoC, AD-S1298 AFE, BLE 5.0 connectivity, LiPo battery (≥ 48 h)
2. **Mobile Application:** iOS/Android app for real-time ECG display, data relay, symptom logging
3. **Cloud Backend:** Secure data processing, storage, AI/ML inference engine
4. **AI Module:** CNN-LSTM model for automated arrhythmia detection (11 classes)
5. **Clinical Dashboard:** Web-based interface for physician review and report generation

3.2 Principles of Operation

The device acquires surface ECG signals via Ag/AgCl electrodes, digitizes them at 500 Hz with 24-bit resolution, applies digital filtering, and transmits data via BLE to a mobile device. The mobile app relays data to the cloud backend via HTTPS/TLS 1.3. The AI module analyzes ECG segments using a CNN-LSTM architecture to detect and classify arrhythmias. Results are presented on the clinical dashboard for physician review.

Section 4: Substantial Equivalence Discussion

4.1 Comparison Table

Feature	Horizon HM-H100 (Subject)	Zio XT (Predicate K142497)	Difference Raises New Q?
Intended Use	Ambulatory ECG monitoring + arrhythmia detection	Ambulatory ECG monitoring + arrhythmia detection	No — Same
Patient Contact	Ag/AgCl electrodes on chest	Ag/AgCl electrodes on chest	No — Same
Number of Channels	3 channels	1 channel	No — More channels provide more diagnostic information
Recording Duration	Up to 48 hours	Up to 14 days	No — Shorter duration, same diagnostic purpose
Data Transmission	Real-time via BLE + cloud	Store and forward (mail-in device)	No — Real-time provides earlier detection
Arrhythmia Analysis	AI/ML (CNN-LSTM)	Proprietary algorithm	No — Both use computerized analysis
Power Source	Rechargeable LiPo	Non-rechargeable battery	No — Same function
Biocompatibility	Ag/AgCl, ABS/PC housing	Ag/AgCl, medical adhesive	No — Same material class
Prescription Use	Yes	Yes	No — Same
Patient Population	Adults ≥18 years	Adults ≥18 years	No — Same

4.2 Technological Differences and Justification

Difference 1: Real-time BLE connectivity (vs. store-and-forward)

- This represents an improvement that enables earlier detection of critical arrhythmias
- BLE communication is a well-established technology with known safety profile
- Does not raise new safety or effectiveness questions

Difference 2: AI/ML algorithm (CNN-LSTM vs. proprietary)

- Both devices use computerized arrhythmia detection algorithms
- The CNN-LSTM architecture has been validated to achieve performance comparable to or exceeding the predicate's algorithm (see Performance Data section)

- The AI module functions as a clinical decision support tool, same as the predicate
- Comprehensive AI/ML documentation provided per FDA guidance

Difference 3: 3-channel ECG (vs. 1-channel)

- Additional channels provide more diagnostic information
- This is a clinical improvement, not a new safety concern
- 3-channel Holter is the traditional standard configuration

4.3 Conclusion

The Horizon Medical Holter ECG System is **substantially equivalent** to the predicate device (Zio XT, K142497). The identified technological differences do not raise new questions of safety or effectiveness, as demonstrated by the non-clinical and clinical data presented in this submission.

Section 5: Performance Data

5.1 Non-Clinical (Bench) Testing

Test	Standard	Result
Electrical Safety	IEC 60601-1:2020	PASS — Report #[XXX]
EMC	IEC 60601-1-2:2014+AMD1	PASS — Report #[XXX]
Holter-specific	IEC 60601-2-47:2012+AMD1	PASS — Report #[XXX]
ECG Accuracy	AHA/AAMI EC38	Meets all accuracy requirements
Battery Safety	IEC 62133-2:2017	PASS — Report #[XXX]
Biocompatibility	ISO 10993-5, -10	PASS — Report #[XXX]
BLE Performance	Bluetooth SIG	PASS — Range $\geq 10\text{m}$, stable throughput

5.2 Software Testing Summary

Test Level	Test Cases	Pass	Fail	Coverage
Unit Testing	[N]	[N]	0	[%]
Integration Testing	[N]	[N]	0	—
System Testing	[N]	[N]	0	—
Regression Testing	[N]	[N]	0	—

5.3 AI/ML Algorithm Performance

Arrhythmia	Sensitivity	Specificity	AUC-ROC
Atrial Fibrillation	[≥95%]	[≥95%]	[≥0.97]
Ventricular Tachycardia	[≥95%]	[≥98%]	[≥0.98]
Bradycardia	[≥90%]	[≥95%]	[≥0.95]
PVC	[≥85%]	[≥95%]	[≥0.93]

[Full AI/ML documentation provided as separate exhibit]

5.4 Usability Testing Summary

Population	N	Critical Use Errors	Task Completion
Patients	[N]	0	[≥95%]
Clinicians (dashboard)	[N]	0	[≥95%]
Technicians (placement)	[N]	0	[100%]

Section 6: Labeling

6.1 Instructions for Use (Summary)

The complete Instructions for Use (IFU) are provided as Exhibit [X] and include:

- Device description and components
- Indications for use and contraindications
- Warnings and precautions
- Patient preparation and electrode placement instructions
- Mobile app setup and operation
- Troubleshooting guide
- Technical specifications
- Explanation of AI/ML functionality and limitations
- Symbol glossary (ISO 15223-1)
- Maintenance and disposal instructions

6.2 Key Warnings and Precautions

- **Rx Only** — Federal law restricts this device to sale by or on the order of a physician
- The AI-based arrhythmia detection is intended as a decision support tool only. All results must be reviewed and confirmed by a qualified healthcare professional.
- Not intended for real-time patient monitoring in acute care settings (ICU, ER)

- Not validated for pediatric patients (<18 years)
- Patients with implanted electronic devices should consult their physician before use

Section 7: Standards and Declarations of Conformity

Standard	Title	Declaration
IEC 60601-1:2020	General safety and essential performance	Full conformity declared
IEC 60601-1-2:2014+AMD1	EMC requirements	Full conformity declared
IEC 60601-1-6:2010+AMD1+AMD 2	Usability	Full conformity declared
IEC 60601-1-11:2015+AMD1	Home healthcare environment	Full conformity declared
IEC 60601-2-47:2012+AMD1	Ambulatory ECG	Full conformity declared
IEC 62304:2006+AMD1	Medical device software life-cycle	Full conformity declared
IEC 62366-1:2015+AMD1	Usability engineering	Full conformity declared
ISO 10993-1:2018	Biocompatibility evaluation	Full conformity declared
ISO 14971:2019	Risk management	Full conformity declared

Section 8: Truthful and Accuracy Statement

I certify that, in my capacity as [Title] of [Company Name], this submission includes all information, published or known to me, that is relevant to the evaluation of safety and effectiveness of the device. I further certify that no material fact has been omitted, and that the information presented is truthful and accurate to the best of my knowledge.

Signature: _____

Name: [Name]

Title: [Title]

Date: [Date]

Exhibits List

Exhibit	Document	Pages
A	Indications for Use (FDA Form 3881)	1
B	Device Description and Specifications	[N]
C	Substantial Equivalence Comparison	[N]
D	Electrical Safety Test Report (IEC 60601-1)	[N]
E	EMC Test Report (IEC 60601-1-2)	[N]
F	Holter ECG Test Report (IEC 60601-2-47)	[N]
G	Biocompatibility Evaluation (ISO 10993)	[N]
H	Battery Safety Test Report (IEC 62133-2)	[N]
I	Software Documentation (IEC 62304)	[N]
J	AI/ML Documentation	[N]
K	Cybersecurity Documentation	[N]
L	SBOM	[N]
M	Risk Management Summary (ISO 14971)	[N]
N	Usability Summary (IEC 62366-1)	[N]
O	Clinical Performance Data	[N]
P	Labeling (IFU)	[N]
Q	Standards Declarations of Conformity	[N]
R	Financial Certification (FDA Form 3674)	[N]

Control de Versiones:

Versión	Fecha	Autor	Descripción
1.0	2026-05-16	Regulatory Affairs	Creación inicial
